

Name: _____

Class: _____

Friday, December 1, 2023

Test

Please use a pencil to complete the test. Print your name on the Name line above. Read the instructions for each section carefully. When you have completed the test, place your test face down on your desk and raise your hand.

- 1. Communication between a computer and a keyboard involves _____
1 transmission.

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see Figure 1.2a). Keyboards and traditional monitors are examples of simplex devices.

- | | |
|--|---------------------------------------|
| ☐ (A) Automatic | ☐ (B) Full-duplex |
| ☒ (C) Simplex | ☐ (D) Half-duplex |

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-basics/>

- 2. Communication between two devices can be simplex, half-duplex, or full-duplex as shown in
1 Figure 1.2.

Which of the following isn't a type of transmission mode?

- | | |
|---|---------------------------------------|
| ☒ (A) physical | ☐ (B) simplex |
| ☐ (C) full duplex | ☐ (D) half duplex |

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-transmission-modes/>

- 3. A transmission that generally involves dedicated circuits.
1

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see Figure 1.2a).

- | | |
|--|---------------------------------------|
| ☒ (A) simplex | ☐ (B) half duplex |
| ☐ (C) full duplex | ☐ (D) semi-duplex |

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-transmission-modes/>

- 4. A transmission mode that can transmit data in both the directions but transmits in
1 only one direction at a time.

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (see Figure 1.2b).

- ☐ (A) simplex ☒ (B) half duplex
☐ (C) full duplex ☐ (D) semi-duplex

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-transmission-modes/>

- 5. Telephone networks operate in this mode.
1

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

- ☐ (A) simplex ☐ (B) half duplex
☒ (C) full duplex ☐ (D) semi-duplex

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-transmission-modes/>

- 6. A walkie-talkie operates in _____
1

half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

- ☐ (A) simplex ☒ (B) half duplex
☐ (C) full duplex ☐ (D) semi-duplex

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-transmission-modes/>

- 1 7. Data can flow only in one direction all of the times in a _____ mode.
In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see Figure 1.2a).

- ☒ (A) simplex ☐ (B) half-duplex
☐ (C) full-duplex ☐ (D) None of the choices are correct

Notes:

https://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html

- 1 8. Data can flow only in both direction all of the times in a _____ mode.
The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction.

- ☐ (A) simplex ☐ (B) half-duplex
☒ (C) full-duplex ☐ (D) None of the choices are correct

Notes:

https://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html

- 1 9. Both stations can transmit and receive data simultaneously in
The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction.

- ☐ (A) simplex mode ☐ (B) half duplex mode
☒ (C) full duplex mode ☐ (D) unicode

Notes: <https://mcqslearn.com/cs/computer-networks/quiz/quiz-questions-and-answers.php?page=20>

- 10. A.....is the physical path over which a message travels.

1

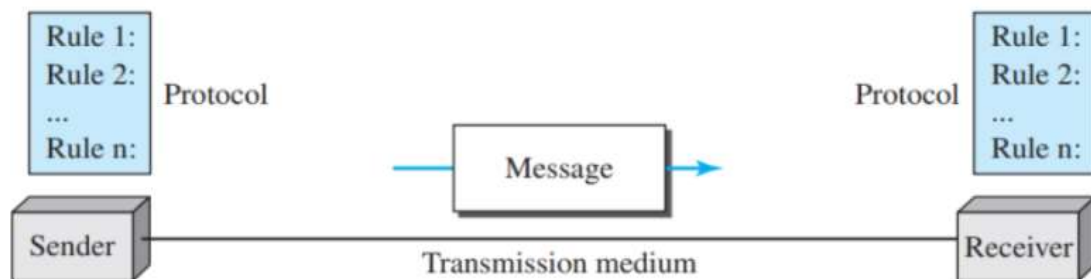
Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver

- (A) path (B) medium
(C) protocol (D) route

- 11. Which of these element is not involved in the process of communication?

1

Five components of data communication



- (A) pipe (B) sender
(C) message (D) channel

- 12. A.....set of rules that governs data communication

1

Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese

- (A) protocols (B) standards
(C) RFCs (D) Servers

- 13. Ais the person who transmits a message.

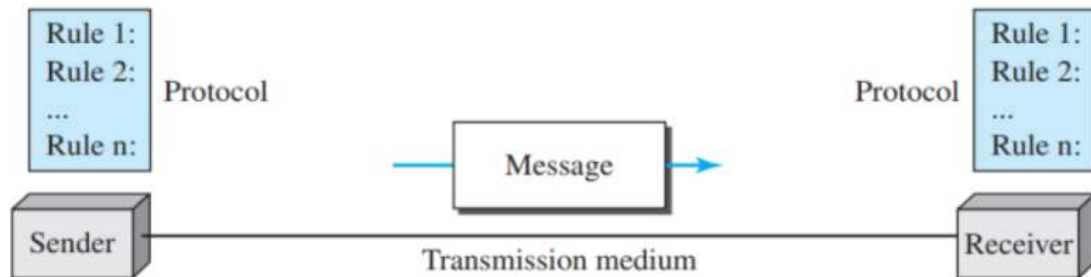
1

sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera.

- (A) Receiver (B) sender
(C) Message (D) Transmission medium. T

- 14. Which of the following is not one of the components of a data communication system?
1

Five components of data communication



- (A) Sender (B) messege
(C) meduim (D) All of the choices are correct

- 15. The function of the data transmission element is.....
1

Data communications are the exchange of data between two devices

- (A) to transfer data from one element to another (B) to Modify the data
(C) to process the data (D) to separate the signal hidden in the noise

- 16. Any set of digits or alphabets are generally referred as _____
1

- (A) Characters (B) Symbols
(C) Bits (D) Bytes

- 17. what is a pixel?
1

Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements),

- (A) pixel is the element of a digital image (B) pixel is the element of an analog image
(C) pixel is the cluster of a digital image (D) pixel is the cluster of an analog image

— 18. Which color model is used for most computer model and video systems?

1

There are several methods to represent color images. One method is called RGB, so called because each color is made of a combination of three primary colors: red, green, and blue.

☒ A RGB

☐ B RYB

☐ C CMU

☐ D HSV

— 19. what does ASCII stand for?

1

The American Standard Code for Information Interchange (ASCII),

☒ A American Standard Code for Information Interchange

☐ B American Scientific Code for information interchange

☐ C American Scientific Code for interchanging information

☐ D American Standard Code for interchanging information

— 20. The largest geographic area a wide area network (WAN) can span is _____.

1

a WAN has a wider geographical span, spanning a town, a state, a country, or even the world.

☐ A a town

☐ B a state

☐ C a country

☒ D the world

Notes:

https://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html

— 21. Data communication system spanning states, countries, or the whole world is

1

a WAN has a wider geographical span, spanning a town,
a state, a country, or even the world.

☐ A LAN

☒ B WAN

☐ C MAN

☐ D PAN

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-topology/>

— 22. WAN stands for _____
1

A wide area network (WAN) is also an interconnection of devices capable of communication. However, there are some differences between a LAN and a WAN.

- (A) World area network (B) Wide area network
(C) Web area network (D) Web access network

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-topology/>

— 23. What is the purpose of a WAN?
1

a WAN has a wider geographical span, spanning a town, a state, a country, or even the world.

- (A) To allow computers to communicate with each other in the same area (B) To allow computers to communicate with each other - anywhere in the world
(C) To allow computers to communicate with each other in the same country (D) To allow computers to communicate with each other in the home

Notes: <https://quizizz.com/admin/quiz/5dbf6c102af6a1001a2e2b2c/lan-man-wan>

— 24. The Word Wide Web (WWW) is the largest ...
1

- (A) Local Area Network (LAN) (B) Wide Area Network (WAN)
(C) Metropolitan Area Network (MAN)

Notes: <https://quizizz.com/admin/quiz/5dbf6c102af6a1001a2e2b2c/lan-man-wan>

— 25. Wide Area Networks (WANs)

1

A wide area network (WAN) is also an interconnection of devices capable of communication. However, there are some differences between a LAN and a WAN. A LAN is normally limited in size, spanning an office, a building, or a campus; a WAN has a wider geographical span, spanning a town, a state, a country, or even the world. A LAN interconnects hosts; a WAN interconnects connecting devices such as switches, routers, or modems. A LAN is normally privately owned by the organization that uses it; a WAN is normally created and run by communication companies and leased by an organization that uses it. We see two distinct examples of WANs today: point-to-point WANs and switched WANs

- ☐ (A) cover a large geographical area ☐ (B) use cables, telephone lines, satellites and radio waves to connect
- ☐ (C) are used by international organisations ☒ (D) All 3 options

Notes: <https://quizizz.com/admin/quiz/5ebea84ae65fe6001bb442fd/networks-topologies?fromSearch=true&source=null>

— 26. How is software reliability defined?

1

In addition to accuracy of delivery, network **reliability** is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

- ☒ (A) time ☐ (B) efficiency
- ☐ (C) quality ☐ (D) speed

Notes: <https://www.sanfoundry.com/software-engg-mcqs-software-reliability/>

Question:6

— 27. A local area network (LAN) is defined by.....
1

! We use a few criteria such as size, geographical coverage, and ownership

- ☒ A the geometric size of the network ☐ B the maximum number of hosts in the network
☐ C the maximum number of hosts in the network and/or the geometric size of the network ☐ D the topology of the network

Notes:

https://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html

Question:1

— 28. WLAN stands for.....
1

A local area network (LAN)

- ☒ A Wireless Local Area Network ☐ B Wire Lost Area Network
☐ C Wireless Local Ambiguity Network ☐ D Wired Local Area Network

Notes: <https://testbook.com/objective-questions/mcq-on-lan-technologies--5eea6a0939140f30f369d915>

Question:9

— 29. Which of the following is a type of wireless communication?
1

! two types of networks, LANs and WANs,

- ☐ A LAN ☐ B WAN
☐ C PAN ☒ D All of the mentioned

Notes: <https://www.sanfoundry.com/1000-wireless-mobile-communications-questions-answers/>

Question:3

— 30. Which of the following is not one of the network criteria?

1

A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

- ☐ (A) Performance ☐ (B) Reliability
☐ (C) Security ☒ (D) All of the choices are correct

Notes:

https://highered.mheducation.com/sites/0073376221/student_view0/chapter1/quizzes.html

Question:8

— 31. Which of the following is a small single site network?

1

A local area network (LAN) is usually privately owned and connects some hosts in a single office, building, or campus. Depending on the needs of an organization, a LAN can be as simple as two PCs and a printer in someone's home office, or it can extend throughout a company and include audio and video devices. Each host in a LAN has an

- ☒ (A) LAN ☐ (B) DSL
☐ (C) MAN ☐ (D) WAN

Notes: <https://testbook.com/objective-questions/mcq-on-lan-technologies--5eea6a0939140f30f369d915>

Question:14

— 32. A term that refers to the way in which the nodes of a network are linked together

1

The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices

- ☐ (A) network ☒ (B) topology
☐ (C) connection ☐ (D) interconnection

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-network-topologies/>

— 33. three or more devices share a link in _____ connection.

1

Multipoint A multipoint connection is one in which more than two specific devices share a single link

- | | |
|--|---|
| ☐ (A) unipoint | ☒ (B) multipoint |
| ☐ (C) point to point | ☐ (D) simplex |

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-basics/>

— 34. _____ topology requires a multiple connection

1

The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.

- | | |
|--------------------------------|--|
| ☐ (A) star | ☐ (B) mesh |
| ☐ (C) ring | ☒ (D) bus |

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-topology/>

— 35. the participating computers in a network are referred to as:

1

The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

- | | |
|--|-----------------------------------|
| ☐ (A) clients | ☐ (B) servers |
| ☒ (C) nodes | ☐ (D) CPUs |

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-network-topologies/>

- 36. a topology that is responsible for describing the geometric arrangement of
1 components that make up the lan.

The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices.

- ☐ (A) complex ☒ (B) physical
☐ (C) logical ☐ (D) incremental

Notes: <https://www.sanfoundry.com/computer-fundamentals-questions-answers-network-topologies/>

- 37. _____ topology requires a multipoint connection.
1

- ☐ (A) Star ☐ (B) Mesh
☐ (C) Ring ☒ (D) Bus

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-topology/>

- 38. Which network topology requires a central controller or hub?
1

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.

- ☒ (A) Star ☐ (B) Mesh
☐ (C) Ring ☐ (D) Bus

Notes: <https://www.sanfoundry.com/computer-networks-mcqs-topology/>